

PROPERTYHUB TECHNICAL REPORT

1. Project Overview

PropertyHub is a web-based real estate platform that connects landlords, agents, tenants, and buyers.

The platform allows landlords to list properties for rent or sale, agents to repost approved listings, and tenants/buyers to search and inquire about properties.

The system aims to simplify property management and property discovery through a centralized digital platform.

2. Objectives

Main Objectives

- Provide a platform for property listings.
 - Allow landlords to manage properties.
 - Allow agents to market properties.
 - Enable tenants to search for rental properties.
 - Enable buyers to search for properties for sale.
 - Provide secure authentication and authorization.
 - Create a responsive user-friendly interface.
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3. User Roles

Landlord

Permissions:

- Register account
 - Login
 - Create profile
 - Post properties
 - Upload property images
 - Edit properties
 - Delete properties
 - View inquiries
 - Approve agent repost requests
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Agent

Permissions:

- Register account
 - Login
 - Create profile
 - Request listing repost rights
 - Repost approved properties
 - Receive inquiries
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Tenant

Permissions:

- Register account
 - Login
 - Search rental properties
 - Save favorite properties
 - Request inspections
 - Contact landlords
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Buyer

Permissions:

- Register account
 - Login
 - Search properties for sale
 - Save favorite properties
 - Schedule inspections
 - Contact agents or landlords
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Admin

Permissions:

- Manage users
- Manage properties
- Verify agents
- Remove fake listings
- Suspend accounts

- Monitor platform activities
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4. Functional Requirements

Authentication

- User registration
- User login
- JWT authentication
- Password hashing

Property Management

- Create property
- Edit property
- Delete property
- Upload images

Search System

- Search by location
- Search by property type
- Search by price range
- Search by rent/sale

Agent Repost System

- Request permission
- Landlord approval
- Repost listing

Favorites

- Save property
- Remove property

Inspection Requests

- Schedule inspections
 - Approve inspections
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5. Non-Functional Requirements

Performance

- Fast loading pages
- Optimized database queries

Security

- JWT Authentication
- Password Hashing
- Role Based Access Control

Scalability

- Modular backend architecture
- RESTful API design

Availability

- Cloud deployment
 - Database backup support
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6. Technology Stack

Frontend

- HTML5
- CSS3
- JavaScript (Vanilla)

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas

Authentication

- JWT
- bcryptjs

File Upload

- Multer
- Cloudinary

Deployment

- Frontend → Vercel
 - Backend → Render
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7. Database Design

Users

- id
- name
- email
- password
- role
- phone
- profileImage
- createdAt

Properties

- id
- title
- description
- type
- listingType
- price
- location
- bedrooms
- bathrooms
- images
- landlordId
- status

Reposts

- id
- propertyId
- landlordId
- agentId
- commission
- approved

Favorites

- id
- userId
- propertyId

InspectionRequests

- id
 - propertyId
 - userId
 - inspectionDate
 - status
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8. Future Improvements

- Online property payments
 - Real-time chat
 - Google Maps integration
 - Property verification
 - Mobile application
 - AI property recommendations
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9. Expected Outcome

A fully functional property marketplace that enables landlords, agents, tenants, and buyers to interact securely while managing property transactions efficiently.